

AMS1117-3.3

AMS1117-3.3 — 1 A Low Dropout Voltage Regulator

Manufacturer: Advanced Monolithic Systems Package: SOT-223

General Description

The AMS1117-3.3 is a fixed 3.3 V low dropout linear regulator capable of supplying 1 A of output current. Dropout is 1.3 V at full load.

Features

- Fixed output voltage 3.3 V
- Output current up to 1 A
- Low dropout voltage 1.3 V at 1 A
- Input voltage up to 15 V
- Thermal shutdown and current limit

Pin Configuration

Pin assignment for the SOT-223 package (top view).

Pin	Name	Function
1	GND	Ground
2	VOUT	Regulated 3.3 V output
3	VIN	Supply input
Tab	VOUT	Tab is connected to VOUT

Absolute Maximum Ratings

Stresses beyond these values may cause permanent damage to the device.

Symbol	Parameter	Value	Unit
VIN	Maximum input voltage	15	V
IOUT	Output current	1.0	A
Ptot	Power dissipation	Internally limited	-
Tj	Operating junction temperature	125	degC
Tstg	Storage temperature	-65 to 150	degC

Electrical Characteristics

Typical values at TA = 25 degC unless otherwise noted.

Symbol	Parameter	Test condition	Min	Typ	Max	Unit
VOOUT	Output voltage	VIN = 5 V, IOOUT = 10 mA	3.267	3.300	3.333	V
Vdrop	Dropout voltage	IOOUT = 1 A	-	1.1	1.3	V
Iq	Quiescent current	VIN = 5 V	-	5	10	mA
PSRR	Ripple rejection	f = 120 Hz	60	72	-	dB

Thermal Characteristics

Symbol	Parameter	Value	Unit
RthJA	Thermal resistance junction-ambient	90	K/W
RthJC	Thermal resistance junction-case	15	K/W

Package Dimensions

Outline drawing for the SOT-223 package. All dimensions in millimetres.

Dimension	Min	Nom	Max	Unit
A (body length)	6.30	6.50	6.70	mm
B (body width)	3.30	3.50	3.70	mm
C (height)	1.50	1.60	1.80	mm

Ordering Information

Order code	Package	Marking	Packing
AMS1117-3.3	SOT-223	AMS1117 3.3	Reel, 1000 pcs

Document revision 1.0. Values are typical at TA = 25 degC. Verify against the latest datasheet from Advanced Monolithic Systems before design release.